Practical 1

AIM: Define a simple services like Converting Rs into Dollar and Call it from different platform like JAVA and PYTHON.

Steps:

1] Open the NetBeans , and you will get the following screen. Close the startpage.

2] Nowclick on the file tab and click on new project you will get the following screen:

3] In Categories select Java Web and in Projects , Select Web Application .After selecting click onnext . You will get the following window:

4] NowGive name to the Project Name: , and click on next . you will get the following window then again click on finish:

5] You will get the following screen now carefully see in projects section your recently created project appears double click to expand it:

6] NowRight click on the project and select new and then select Web Service ,As shownBelow:

7] after clicking Web Service following window should appear , now give name to web serviceand give package as “server”. As shown in the image:

After clicking finish you should get the following window erase the mentioned code :

8] Nowright click any where and click on insert code and select Add Web Service Operation:

9] the following window would appear: Just give name to the method or operation and click on add button to add parameters to the method as here we are converting dollar to rupees we should need only one parameter .

10] give the name to the variable select data type as double and click on ok as shown below:

11] After clicking ok code will autogenerate , make changes In that code as mentioned below:

12] now our webservice is ready now right click on project and click on deploy:

13] now right click on webservice and click on test web service you will get the following output:

So this is how we created our web service and deployed it.

Creating a java Client using jsp

14] now our web service is successfully deployed. Right click on web pages and select new and select jsp as shown below:

15] give name and click on finish as shown below do it 2 times on for input and one foroutput:

16] In input.jsp create a form for taking user input as shownbelow:

In this code set action = the jsp file where u want output and give name to textbox input.
17] now we have to create web service client, for same right click on project and select new then select web service client you will get the following screen:

At this step click on browse and select your project and click ok after clicking ok you will get wsdl url as shown below:

At this stage copy the url and paste it in notepad for future.

Give package name as "Client".

Now click on finish.

18] now in project section you can se a new folder is been created named "web servicerefrence". double click to expand it you should get the following

Hold the operation or your operation name and drag it into the output.jsp in body tag as shown : You will get the following auto generated code:

19] now makechangesas shown below in thecode:

20] now Deploy our project again and right click on input.jsp and click run file you will redirect to a browser page as shown :

Enter any numerical value in text box and click on submit you will get the following output:

Python client:

1] for consuming java web services in python we should repeat the above steps till step np.19 . Note: for getting output the project or web service should be deployed . After doing all of the steps write the following code in python idle:

from zeep import Client wsdl_url='http://localhost:8080/pract/Currency?wsdl' client=Client(wsdl_url) #call the method on the webservice d=float(input("Enter Currency in Indian Rupees:")) result = client.service.InrtoDollar(d) print(result)

2] replace wsdl_url with the url which was copied at the time of web service client creation .

4] after running the above code you should get the following output:

Practical 2

Aim: Create a Simple SOAP service.

Steps:

1] Open the NetBeans , and you will get the following screen. Close the startpage.

2] Nowclick on the file tab and click on new project you will get the following screen:

3] In Categories select Java Web and in Projects , Select Web Application .After selecting click onnext . You will get the following window:

4] NowGive name to the Project Name: , and click on next . you will get the following window then again click on finish:

5] You will get the following screen now carefully see in projects section your recently created project appears double click to expand it:

6] NowRight click on the project and select new and then select Web Service ,As shownBelow:

7] after clicking Web Service following window should appear , now give name to web serviceand give package as “server”. As shown in the image:

After clicking finish you should get the following window erase the mentioned code :

8] Nowright click any where and click on insert code and select Add Web Service Operation:

9] the following window would appear: Just give name to the method or operation and click on add button to add parameters to the method as here we are converting dollar to rupees we should need only one parameter .

10] give the name to the variable select data type as double and click on ok as shown below:

11] After clicking ok code will autogenerate , make changes In that code as mentioned below:

12] now our webservice is ready now right click on project and click on deploy:

13] now right click on webservice and click on test web service you will get the following output:

So this is how we created our web service and deployed it.

14] Now our web service is successfully deployed. Right click on web pages and select new and select jsp as shown below:

Practical 3

Aim: Create a Simple REST Service to demonstrate CRUD Operations with "Student" database.

Steps:

1.Click on Window menu and click on Projects, Files & Services to open it.

2.Right click on Java DB and then click on Start Server to start the server .

3.Now expand Java DB and right click on sample and then click on connect to connect the sample database with server.
4.Now create a web application with the name CRUD_Operation. A window will open like following pic.

5.Create an entity class. Right click on project name -> New -> Entity Class.

6.A window will appear like bellow pic. Enter following data and click on Next …. Class Name ->Stud_Name Package Name -> p1

7.Click on finish.

8.Right click on project name and create JSF Pages from Entity Classes.

9.Select and click on Add button and then Next button on below

10.A window like below will appear on the screen. Enter the data into that window as entered in below pic and click on Next button.

11.Now click on Finish.

12.Right click on project name and create RESTful Web Services from Entity Classes.

13.Repeat step 9 and then it will go on next page. Then enter the p1.service in Resource Package and then click on Finish button.

14Now open Stud_Name.java file under p1 package.

15.In this file at line number 24, do the right click and select Insert Code.

16.A new list will appear. Click on Add Property

17.A new window will open. Enter name as studName. Make sure name should be exact same as of mine and then click on OK button. Actually we are setting getter and setter method for studName.

18.Now right click on web application name and deploy it.

19.Now right click on project name and run it

20.A window will open in browser like below….

Practical 4

AIM: Develop application to consume Google's search / Google's Map RESTful Web service.

Steps:

- Develop application to consume Google's search / Google's Map RESTful Web service.

1. First, we need to create a Java Web Application with any name, let it be GoogleMap here using Netbeans IDE.
2. The code inside the input.jsp will be similar to this input.jsp Input.jsp

```
<%@pagecontentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>

<html>

<head>

<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">

<title>JSP Page</title>

</head>

<body>

<form action="index.jsp">

<pre>

Enter latitude:<input type="text" name="t1" />

Enter longitude:<input type="text" name="t2" />

<input type="submit" value="Show" />

</pre>

</form>

</body>

</html>
```

3.Before running the application, we need the Google API key. The steps are shown here:- Visit Google APIs Console (https://console.developers.google.com, you have to login with your Google account).

4.Before running the application, we need the Google API key. The steps are shown here:- Visit Google APIs Console (https://console.developers.google.com, you have to login with your Google account).

Follow these steps

1. Firstly Go to Google Maps - https://console.developers.google.com
2. Sign In with your Google account and agree to the terms and conditions
3. Create a project and name it as you want
4. Select the project and click on it
5. Navigate to APIs and Services and click on Libraries
6. Enable the following Libraries one by one
    1. Google Javascript API
    2. Google Place API
    3. Google Direction API
    4. Google Geolocation API
    5. Google Geocoding API
7. Navigate to APIs, Services, and click on Credentials
8. Click on Create Credentials and then the API key and it will show up a screen with API Key
9. Now you can optionally restrict the API key.

Almost done!

5. Create another file index.jsp

Index.jsp

```
<%@pagecontentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<style> #map{
height: 400px; width:
100%;
}
```

```
</style>
</head>
<body>
<%
double lati=Double.parseDouble(request.getParameter("t1")); double
longi=Double.parseDouble(request.getParameter("t2"));
%>
<h3> Google Maps</h3>
<div id="map"></div> <script
lang="javascript"> function
initMap() {
var info={lat: <%=lati%>, lng: <%=longi%>}; var map
=newgoogle.maps.Map(document.getElementById('map'),
{ zoom: 4, center:
info
});
var marker = new google.maps.Marker({ position:
info,
map: map
});
}
</script>
<script async defer src="put
your key here">
</script>
</body>
</html>
```

Practical 5

Aim: Installation and Configuration of virtualization using KVM.

Steps to install Virtualization using KVM

To create a virtual machine first we need to ensure that virtualization is enabled on our system. It is mandatory to create virtual machines. There are multiple ways to check if virtualization is enabled,

$ egrep -c '(vmx|svm)' /proc/cpuinfo

The above command /proc/cpuinfo gives information about the processor. The output of the command will be a number. The output number 1 or more than that represents that virtualization is enabled, output 0 says you need to enable virtualization on your system.

Virtualization Inability

$ lscpu | grep Virtualization

This command is used to check which type of virtualization your processor supports. If the system contains a CPU with Intel VT support, the above command will provide the following output

Virtualization Type

Installing KVM on Ubuntu

$ sudo apt-get install virt-manager

Now we know our system is capable of creating and running virtual machines, it's time to install tools which will create our virtual machines. To install KVM and other KVM dependencies such as virt-manager, bridge-utils enter command:

$ sudo apt -y install bridge-utils cpu-checker libvirt-clients libvirt-daemon qemu qemu-kvm

1. bridge-utils: The bridge-utils package contains a utility needed to create and manage bridge devices. This is useful in setting up networks for a hosted virtual machine (VM).
2. cpu-checker: Outputs the specifications of CPU
3. libvirt-clients: a toolkit to manage virtualization platforms/clients and hypervisors
4. qemu: A program that can run the operating system of the machine on different machines
5. qemu-kvm: Runs process using KVM module

All dependencies are installed now run command its time to check if KVM ins installed successfully:

$ sudo kvm-ok

Check if KVM is Installed properly

Also, we need to confirm if the virtualization daemon – libvritd-daemon – is running, to do so enter the command.

$ sudo systemctl status libvirtd

If the output is not active: running you need to start daemon thread

If the daemon thread is not running enter the following command to start the thread,

$ sudo systemctl enable –now libvirtd

Adding a user to KVM

In this part, we are going to create a user for KVM. To prevent root user from using KVM and root user is only available when root user is a part of libvirt/libvirtd group. To add a new user to KVM, use the following command, *sudo adduser [username] libvirtd*

[username] enter the username of your choice if the output is

Adding User to KVM

If your KVM is already a member of the non-root user and serves the same purpose as libvirtd group then you don't need to add yourself to the group.

Creating Virtual Machine

There are two ways to create a virtual maclsne

1. Using the command line
2. Using the graphical interface

- **Create a Virtual Machine via Command Line**

virt-install is a command which is used to create virtual machines in Linux, following is the command which creates a VM.

sudo virt-install –name=ubuntu-guest –os-variant=ubuntu20.04 –vcpu=2 –ram=2048 –graphics none –location=[local path to iso file] –network bridge:vibr0

The above command creates a Ubuntu virtual machine with version 20.04 and the name ubuntu-guest.

1. Name: specify the name of the virtual machine being created
2. vcpu: Number of virtual CPUs to configure for the guest.
3. Ram: Memory to allocate for guest instance in megabytes. According to your machine, you can specify the given memory of the VM.

4. Graphics spice: If no graphics option is specified, “virt-install” will default to ‘–graphics vnc’ if the display environment variable is set, otherwise ‘–graphics none’ is used.
5. Location: location of iso file on which virtual machine will be built. It can be the path to an ISO image, or to a CDROM device. It can also be a URL from which to fetch/access a minimal boot ISO image.
6. Network-bridge: Connect the guest to the host network, Connect to a bridge device in the host called “BRIDGE”.

Create a Virtual machine using a graphical interface

If you are not much familiar with the command line don’t worry there’s another way to create a virtual machine using a tool called virt-manager you can easily create virtual machines. Steps to create VM using virt-manager,

Step 1: Launch virt-manager

Step 2: Choose installation media it can be an iso file of OS, you can install from the network or can be a disk copy

Step 3: Enter the path/URL of your file click next

Step 4: Enter memory and CPU requirements

Step 5: Enter required disk space

Step 6: Enter the name of VM and check specifications and hit the finish

Now the installations should be completed and the virtual machine should start running.

Practical 6

AIM: Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure) Infrastructure as a Service (IaaS), Storage.

Steps:

1.Start VMWARE workstation player 15 □ select Open virtual machine then□ Browse a foss cloud iso file□ then select Operating System: Other Version: Other, Name: FOSS.

2.Then click on edit virtual machine and select Processor: 2 or 3, Memory: expand to 165 GB.

3.Once the foss Cloud is launched select "Foss-Cloud installer :default boot options" and press Enter

4.Reboot your system and select foss cloud

5.Types yes to start the installation

6.Select "Demo System" and press Enter

7.Enter the Device name as "sda" and press enter

8.Put credential of foss cloud and start foss cloud 9.Then

Execute below mention command fc-node-

configuration –n demo-system –password admin

10.Then type ifconfig command to get the ip address

11.Then go to any web browser in windows OS and type the IP Address □ FOSS cloud web page will open and you will find login details.

12.After putting the login details you will get the below mention page

Practical 7

AIM: Using AWS Flow Framework develop applications that include a simple workflow. Workflow calls an activity to print hello world to the console. It must define the basic usage of AWS Flow Framework, including defining contracts, implementation of activities and workflow coordination logic and worker programs to host them.

Step 1: Open Terminal and Update and Upgrade your system by command sudo apt-get update && sudo apt-get upgrade

Step 2: Download awscliv2.zip with command curl

"https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip" Step

3: Download awscliv2.sig file with command curl -o awscliv2.sig

https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip.sig Step4: Verify

installation sam --version

Step 5: Type command pip3 install aws-sam-cli

Step 6: Type command sam init in terminal to launch Sam

CLISelect 1st option to use AWS Quick Start Templates

Step 7: Select Template no.1 Hello World Example

Step 8: Type "N" if it ask to use most popular runtime and package type

Open new terminal by pressing ctrl+shift+T and check for python version by command python –version

Select the option according to your python version in my case its option 19- python 3.11
Step 9: Select package type as <u>Zip</u>

Step 10: Now choose Yes option everytime it ask .

Give project name as per your preference in my case its <u>sam-app-test</u>
Step 11: Now one folder will be created by your provided project name go into that folder by command cd (folder startname)

After entering the project folder we will invoke the HelloWorldFunction by using command

sam local invoke "HelloWorldFunction"

It should give you StatusCode:200

Use command sudo snap install docker if docker error occurs.

Command to install docker

1. sudo apt update

2. sudo apt install -y docker.io
3. sudo systemctl start docker
4. sudo systemctl enable docker
5. docker --version

OPTIONAL STEPS IF STILL YOU ARE GETTING ERROR

- sudo usermod -aG docker $USER
- newgrp docker
- docker ps

Step 13: Type command sam local start-api this will give you URL open it in any browser.

Practical 8

AIM: Implementation of Openstack with user and private network creation.

STEPS:

Step 1: Update Ubuntu System

Open the terminal and run the following command to ensure that the system is up to date :

- $ sudo apt update **OR**
- $ sudo apt -y upgrade
- **Sample Output :**

Reboot the system after runn5ng the above command. To reboot the system, run the following command :

- $ sudo reboot

or

- $ init 6

Step 2: Create Stack User

It is important that the devstack must run as a regular user (non-root user) with the sudo enabled.

To keep this note in mind, let's create a new user with the name "stack" and assign the sudo permissions or privileges. To create a stack user, run the following command in your terminal:

- $ sudo useradd -s /bin/bash -d /opt/stack -m stack

Output :

Now, to assign the sudo privileges to the stack user, run the following command :

- $ echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack

Output :

You can switch to the 'stack' user by running the following command:

- $ sudo su - stack

Output :

Step 3: Install the Git

In Most of the Ubuntu systems, git comes by default. But if git is missing on your system, then install it by running the following command:

- $ sudo apt install git -y

Sample Output :

Step 4: Download OpenStack

Once you install the git, use the git command to download the DevStack from Github.

$cd devstack

$cp samples/local.conf .

- $ git clone https://git.openstack.org/openstack-dev/devstack

Output :

Step 5: Create a DevStack Configuration File

First of all, go to the devstack directory by running the following command :

- $ cd devstack

Output :

Now, create a local.conf file in which you have to enter the four passwords and the host IP address :

Output :

Copy the following line of content in the file :

- [[local|localrc]]
-
- # Password for KeyStone, Database, RabbitMQ and Service
- ADMIN_PASSWORD=StrongAdminSecret
- DATABASE_PASSWORD=$ADMIN_PASSWORD
- RABBIT_PASSWORD=$ADMIN_PASSWORD
- SERVICE_PASSWORD=$ADMIN_PASSWORD
-
- # Host IP - To get your Server or VM IP, run the 'ip addr' or 'ifconfig' command
- HOST_IP=192.168.56.103

Output :

Press the ESC, then :wq and then press enter to save and then exit from the local.conf file.

Here, ADMIN_PASSWORD is the password that we will use to log into the OpenStack login page. The default username for an OpenStack is 'admin'.

And HOST_IP is the IP address of your system. To get your Server or VM IP, run the 'ifconfig' or 'ip addr' command.

Step 6 : Install OpenStack with DevStack

To install and run the openstack, execute the following command :

- $ ./stack.sh

DevStack will install the following components:

- Compute Service (Nova) ○ Image Service- Glance ○ Identity Service-Keystone, ○ Block Storage Service - Cinder
- OpenStack Dashboard - Horizon ○ Network Service - Neutron ○ Placement API - Placement ○ Object Storage - Swift

The installation will take about 10-20 minutes, mostly depends on your internet speed.

At the very end of the installation, you will get the host's IP address, URL for managing it and the username and password to handle the administrative task.

Step 7: Accessing OpenStack on a browser

Copy the horizon URL given in the installation output and paste it into your browser : •

http://IP Address/dashboard